Dynamics of a Charged Particle in a Magnetic Field

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Damien RIFFLART & Eliott ARNOLD, Benoît ROESSLINGER

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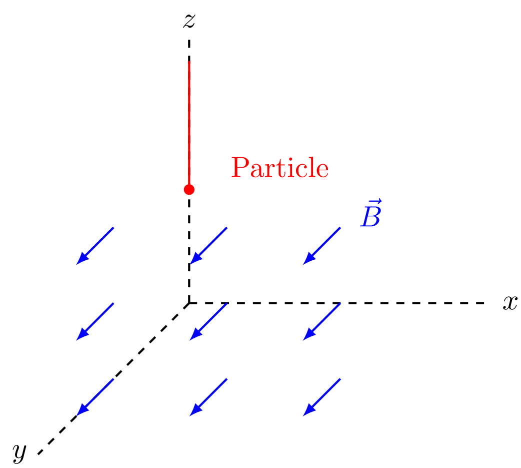
1 Introduction

1.1 Initial Problem

We consider a charged particle entering a uniform magnetic field. Therefore, the particle is subject to the Lorentz force, which deviates its trajectory. **Our goal is to study the motion of a charged particle in a uniform magnetic field, using both classical mechanics and quantum mechanics.** For more information, please visit this website.



1.1.1 Assumptions

In this paper, we make the following assumptions:

- The charged particle has mass m and charge q.
- No electric field is present, and interactions with matter are neglected.

1.2 Experimental Approach

As part of the physics olympiads, our project is to study the motion of electrons placed in a uniform magnetic field. The electrons, resulting from the decay of cosmic muons, are detected using a cloud chamber cooled by nine Peltier modules to approximately -40 degrees Celsius. The experimental setup is not yet fully developed, but we plan to publish the results obtained on the previously mentioned website by 2027.
"Cloud chambers work by creating a supersaturation of alcohol vapor. Vapor trails are then formed when the radiation ionizes the alcohol." (*Instructables, Make a Cloud Chamber Using Peltier Coolers, accessed 23 Dec. 2025*).

1.3 Theoretical Approach

The present article covers both quantum and classical mechanics.
On the one hand, classical mechanics predicts the particle's trajectory merely using the Lorentz force. On the other hand, quantum mechanics introduces a probabilistic interpretation.
We choose to explore both to evaluate their differences.

2 Dynamics of the particle in classical physics

2.1 Hamiltonian and Lorentz force

The Hamiltonian for a particle in a magnetic field is defined as follows:

$$H = \frac{1}{2m}\left[(p_x - \frac{e}{c}A_x)^2 + (p_y - \frac{e}{c}A_y)^2 + (p_z - \frac{e}{c}A_z)^2\right] \tag{1}$$

Using the right-hand rule, we see that the magnetic field in the y direction (B_y) must be nonzero, as we have to deviate the particle on the x plane. The components of the magnetic field are given by

$$B_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}, \quad B_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}, \quad B_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y},$$

Therefore, to obtain a nonzero B_y, either A_z depends on x or A_x depends on z, or both. With this established, the magnetic field exists only along the y-plane. Since $A_y = 0$, we have the following Hamiltonian:

$$H = \frac{1}{2m}\left[(p_x - \frac{e}{c}A_x)^2 + (p_z - \frac{e}{c}A_z)^2 + p_y^2\right] \tag{2}$$

This Hamiltonian is general and holds for any choice of the vector potentials, that we will define in the next parts.

2.2 Moment derivative

2.2.1 In the x-plane

For now, we can choose the vector potentials $A_z = -bx$, $A_x = 0$, and $A_y = 0$. We may now rewrite the Hamiltonian by replacing the vector potentials by their values.

$$H = \frac{1}{2m}\left[p_x^2 + (p_z + \frac{e}{c}bx)^2 + p_y^2\right]$$

Thanks to Hamilton's equation we can easily calculate $\dot{p_x}$:

$$\begin{aligned}
\dot{p_x} &= -\frac{\partial H}{\partial x} = -\frac{\partial}{\partial x}\left[\frac{1}{2m}\left(p_x^2 + (p_z + \frac{e}{c}bx)^2 + p_y^2\right)\right] \\
&= -\frac{1}{2m}\cdot 2\left(p_z + \frac{e}{c}bx\right)\cdot\frac{e}{c}b \\
&= -\frac{eb}{cm}\left(p_z + \frac{e}{c}bx\right) = -\frac{eb}{cm}(mv_z - \frac{eb}{c}x + \frac{eb}{c}x) \\
&= -\frac{eb}{cm}mv_z = -\frac{eb}{c}v_z
\end{aligned}$$

Since $\dot{p_x} = ma_x + \frac{e}{c}\dot{A}_x$ and $\dot{A}_x = 0$, we indeed find the Lorentz force for $v_z = v_0$.

$$\dot{p_x} = ma_x = -\frac{eb}{c}v_z \tag{3}$$

2.2.2 In the z-plane

We modify the vector potential components to:

$$A_z = 0, \quad A_x = bz, \quad A_y = 0.$$

This choice results in the following:

- The magnetic field along the y-plane becomes $B_y = b$.
- The Hamiltonian is now $H = \frac{1}{2m}\left[(p_x - \frac{e}{c}bz)^2 + p_z^2 + p_y^2\right]$
- As you will see in the calculations, $\dot{p_z}$ will be consistent with the Lorentz force.

Using the same method, we find the following.

$$\begin{aligned}
\dot{p_z} &= -\frac{\partial H}{\partial z} = -\frac{\partial}{\partial z}\left[\frac{1}{2m}\left((p_x - \frac{e}{c}bz)^2 + p_z^2 + p_y^2\right)\right] \\
&= -\frac{1}{2m}\cdot 2\left(p_x - \frac{e}{c}bz\right)\cdot\left(-\frac{e}{c}b\right) \\
&= \frac{eb}{cm}\left(p_x - \frac{e}{c}bz\right) = \frac{eb}{cm}(mv_x - \frac{eb}{c}z + \frac{eb}{c}z) \\
&= \frac{eb}{cm}mv_x = \frac{eb}{c}v_x
\end{aligned}$$

Since $\dot{p_z} = ma_z + \frac{e}{c}\dot{A}_z$ and $\dot{A}_z = 0$, we correctly find the Lorentz force.

$$\dot{p_z} = ma_z = \frac{eb}{c}v_x \tag{4}$$

2.2.3 In the y-plane

With the vector potentials chosen as

$$A_z = -bx, \quad A_x = bz, \quad A_y = 0,$$

the Hamiltonian reads $H = \frac{1}{2m}\left[\left(p_x - \frac{e}{c}bz\right)^2 + \left(p_z + \frac{e}{c}bx\right)^2 + p_y^2\right]$.

Since the Hamiltonian does not depend on the coordinate y, $\dot{p}_y$ is conserved $\dot{p}_y = 0$. The velocity component in the y-direction is given by

$$\dot{q}_y = \frac{\partial H}{\partial p_y} = \frac{p_y}{m} = v_y \tag{5}$$

Therefore, the motion along the y-plane is free and unaffected by the magnetic field.

2.3 Position with respect to time

2.3.1 Equations of Motion

In the x-plane

We defined $\dot{p}_x = -\frac{eb}{c} v_z$ in equation (3), and we also have the general expression

$$\dot{p}_x = ma_x + \frac{e}{c}\frac{dA_x}{dt}$$

Equating both expressions with $A_x = 0$, we obtain:

$$ma_x = -\frac{eb}{c} v_z \quad \iff \quad a_x = -\frac{eb}{mc} v_z$$

In the z-plane

We defined $\dot{p}_z = \frac{eb}{c} v_x$ in equation (4). Using the same reasoning as before, with $A_z = 0$, we obtain

$$ma_z = \frac{eb}{c} v_x \quad \iff \quad a_z = \frac{eb}{mc} v_x$$

2.3.2 Position

The accelerations along the x and z directions are coupled through the velocity components:

$$a_x = -\frac{eb}{mc} v_z, \quad a_z = \frac{eb}{mc} v_x$$

Defining the cyclotron frequency as $\omega = \frac{eb}{mc}$, the system can be rewritten as

$$a_x = -\omega v_z, \quad a_z = \omega v_x$$

In the x-plane

Differentiating the first equation with respect to time, and substituting $a_z = \omega v_x$ yields:

$$\dot{a}_x = -\omega \dot{v}_z = -\omega a_z = -\omega^2 v_x$$

Since $\dot{a}_x = \ddot{v}_x$, we obtain the harmonic oscillator equation:

$$\ddot{v}_x + \omega^2 v_x = 0,$$

whose general solution is:

$$v_x(t) = A\cos\omega t + B\sin\omega t$$

For a detailed explanation of why this particular solution is chosen, we wrote one in the Appendix 5.2. We integrate with respect to time, in order to obtain the position of the particle in the x-plane:

$$\begin{aligned} q_x(t) &= \frac{A}{\omega}\sin\omega t - \frac{B}{\omega}\cos\omega t + C \\ &= \frac{v_{x,0}}{\omega}\sin\omega t - \frac{a_{x,0}}{\omega^2}\cos\omega t + \frac{a_{x,0}}{\omega^2} + x_0 \\ &= \frac{v_{x,0}}{\omega}\sin\omega t + \frac{a_{x,0}}{\omega^2}(1-\cos\omega t) + x_0 \\ &= \frac{v_{x,0}}{\omega}\sin\omega t - \frac{v_{z,0}}{\omega}(1-\cos\omega t) + x_0 \quad \text{with} \quad a_{x,0} = -\omega v_{z,0} \end{aligned}$$

Thus, the position along the x-direction is:

$$q_x(t) = \frac{v_{x,0}}{\omega}\sin\omega t - \frac{v_{z,0}}{\omega} + x_0 \tag{6}$$

In the z-plane

Similarly, we have:

$$\dot{a}_z = \omega\dot{v}_x = \omega a_x = -\omega^2 v_z$$

and $\ddot{v}_z + \omega^2 v_z = 0$. Thus, we obtain the velocity in the z-axis:

$$v_z(t) = A\cos\omega t + B\sin\omega t$$

Integrating it with respect to time yields:

$$q_z(t) = \frac{v_{z,0}}{\omega}\sin\omega t + \frac{v_{x,0}}{\omega}(1-\cos\omega t) + z_0 \tag{7}$$

In the y-plane

The motion along the y-axis differs from the other dimensions, as the magnetic field does not affect it. The velocity v_y remains constant. To make this explicit, we denote it as $v_{y,0}$. Integrating with respect to time then yields:

$$q_y = \int v_{y,0}dt = v_{y,0}t + C$$

Using the initial condition $C = y_0$, we find:

$$q_y(t) = v_{y,0}t + y_0 \tag{8}$$

2.4 Conclusion in classical mechanics

To conclude, we have first determined a global Hamiltonian, that holds for every vector potential. When computing $\dot{p}_x$, $\dot{p}_z$ and $\dot{q}_y$, we defined different vector potentials, corresponding to the direction in which we wanted to study the particle's motion. From there, we showed that the particle experiences a Lorentz force causing a circular motion in the (x,z)-plane. In order to address the Lorentz force, we introduced the cyclotron frequency ω. However, the motion in the y-plane stays unaffected by the magnetic field.

The particle's trajectory is thus a **helix**, like a spring stretched along a line. In this case, the particle moves in circles in the (x,z)-plane due to the magnetic force, while also moving at a constant speed along the y-axis.

$$\vec{q}(t) = \begin{pmatrix} \frac{v_{x,0}}{\omega}\sin\omega t - \frac{v_{z,0}}{\omega}(1-\cos\omega t) + x_0 \\ v_{y,0}t + y_0 \\ \frac{v_{z,0}}{\omega}\sin\omega t + \frac{v_{x,0}}{\omega}(1-\cos\omega t) + z_0 \end{pmatrix} \tag{9}$$

3 Dynamics of the particle in quantum physics

3.1 Hamiltonian

In classical mechanics, especially in the Hamiltonian, the dynamics of a system are described using canonical momenta. In quantum mechanics, however, the fundamental objects are operators acting on a Hilbert space. The canonical momentum is thus an operator $\hat{p} = -i\hbar\nabla$, and we obtain the equations of motion using the Schrödinger equation instead of Hamilton's equations. However, we still need the Hamiltonian, which will be a necessary operator to solve Schrödinger's equation. We will use the same Hamiltonian (1) as in classical mechanics, which is :

$$H = \frac{1}{2m}\left[(p_x - \frac{e}{c}A_x)^2 + (p_y - \frac{e}{c}A_y)^2 + (p_z - \frac{e}{c}A_z)^2\right]$$

but with different vector potentials.

3.1.1 Hamiltonian along the x-axis

Using $A_x = 0$, $A_y = 0$ and $A_z = -bx$ yields:

$$H = \frac{1}{2m}\left[\left(p_z + \frac{eb}{c}x\right)^2 + p_x^2 + p_y^2\right]$$

Developing the square of the Hamiltonian gives:

$$\left(p_z + \frac{eb}{c}x\right)^2 = p_z^2 + \frac{e^2b^2}{c^2}x^2 + 2p_z\frac{eb}{c}x,$$

We can remove p_y as the movement along the y-axis is free from the magnetic field. The Hamiltonian becomes:

$$H = \frac{1}{2m}\left(p_z^2 + \frac{e^2b^2}{c^2}x^2 + 2p_z\frac{eb}{c}x + p_x^2\right)$$
$$H = \frac{p_z^2}{2m} + \frac{1}{2m}\frac{e^2b^2}{c^2}x^2 + \frac{eb}{cm}p_z x + \frac{p_x^2}{2m}$$

Substituting the cyclotron frequency $\omega = \frac{eb}{cm}$ defined in the previous section yields:

$$H = \frac{p_z^2}{2m} + \frac{p_x^2}{2m} + \frac{m}{2}\omega^2x^2 + \omega p_z x$$

We can notice two terms in this Hamiltonian: one that we will call H_1, and one that we will call H_2, and define them as $H = H_1 + H_2$

$$H_1 = \frac{m}{2}\omega^2x^2 + \omega p_z x$$
$$H_2 = \frac{p_z^2}{2m} + \frac{p_x^2}{2m}$$

Let's rewrite H_1

$$\begin{aligned}
H_1 &= \frac{m}{2}\omega^2 x^2 + \omega x p_z + \frac{p_z^2}{2m} - \frac{p_z^2}{2m} \\
&= \frac{m}{2}\omega^2 x^2 + \omega x p_z + \frac{m\omega^2}{m\omega^2}\frac{p_z^2}{2m} - \frac{p_z^2}{2m} \\
&= \frac{m}{2}\omega^2 \left(x^2 + \frac{2}{m\omega} x p_z + \frac{p_z^2}{m^2\omega^2} \right) - \frac{p_z^2}{2m} \\
&= \frac{m}{2}\omega^2 (x + \frac{p_z}{m\omega})^2 - \frac{p_z^2}{2m}
\end{aligned}$$

With that value of H_1, the whole Hamiltonian can be rewritten

$$H = H_1 + H_2 = \frac{m}{2}\omega^2 (x + \frac{p_z}{m\omega})^2 - \frac{p_z^2}{2m} + \frac{p_z^2}{2m} + \frac{p_x^2}{2m}$$

Which is equal to

$$H(x) = \frac{p_x^2}{2m} + \frac{m}{2}\omega^2 \left(x + \frac{p_z}{m\omega} \right)^2 \tag{10}$$

It is exactly the equation of a harmonic oscillator with a center of equilibrium which moves depending on the z-position, even if it describes the movement in the x-direction.

3.1.2 Hamiltonian along the z-axis

The steps are the same as in the previous subsection, with the roles of x and z exchanged. We nevertheless detail the calculation for clarity. With different vector potentials $A_x = bz$, $A_y = 0$ and $A_z = 0$, we find:

$$H = \frac{1}{2m} \left[\left(p_x - \frac{eb}{c} z \right)^2 + p_z^2 + p_y^2 \right]$$

Developing the square of the Hamiltonian gives:

$$\left(p_x - \frac{eb}{c} z \right)^2 = p_x^2 + \frac{e^2 b^2}{c^2} z^2 - 2 p_x \frac{eb}{c} z,$$

We can remove p_y as the movement along the y-axis is free from the magnetic field. The Hamiltonian becomes:

$$\begin{aligned}
H &= \frac{1}{2m} \left(p_x^2 + \frac{e^2 b^2}{c^2} z^2 - 2 p_x \frac{eb}{c} z + p_z^2 \right) \\
H &= \frac{p_x^2}{2m} + \frac{1}{2m}\frac{e^2 b^2}{c^2} z^2 - \frac{eb}{cm} p_x z + \frac{p_z^2}{2m}
\end{aligned}$$

Substituting the cyclotron frequency $\omega = \frac{eb}{cm}$ defined in the previous section yields:

$$H = \frac{p_x^2}{2m} + \frac{p_z^2}{2m} + \frac{m}{2}\omega^2 z^2 - \omega p_x z$$

We can notice two terms in this Hamiltonian: one that we will call H_1, and one that we will call H_2, and define them as $H = H_1 + H_2$

$$H_1 = \frac{m}{2}\omega^2 z^2 - \omega p_x z$$
$$H_2 = \frac{p_x^2}{2m} + \frac{p_z^2}{2m}$$

Let's rewrite H_1 so that we make the remarkable identities $(a-b)^2$ appear:

$$\begin{aligned}
H_1 &= \frac{m}{2}\omega^2 z^2 - \omega z p_x + \frac{p_x^2}{2m} - \frac{p_x^2}{2m} \\
&= \frac{m}{2}\omega^2 z^2 - \omega z p_x + \frac{p_x^2}{2m}\frac{m\omega^2}{m\omega^2} - \frac{p_x^2}{2m} \\
&= \frac{m}{2}\omega^2 \left(z^2 - \frac{2}{m\omega} z p_x + \frac{p_x^2}{2m}\frac{2}{m\omega^2}\right) - \frac{p_x^2}{2m} \\
&= \frac{m}{2}\omega^2 \left(z^2 - \frac{2}{m\omega} z p_x + \frac{p_x^2}{m^2\omega^2}\right) - \frac{p_x^2}{2m} \\
&= \frac{m}{2}\omega^2 \left(z - \frac{p_x}{m\omega}\right)^2 - \frac{p_x^2}{2m}
\end{aligned}$$

With that value of H_1, the whole Hamiltonian can be rewritten

$$H = H_1 + H_2 = \frac{m}{2}\omega^2 (z - \frac{p_x}{m\omega})^2 - \frac{p_x^2}{2m} + \frac{p_x^2}{2m} + \frac{p_z^2}{2m}$$

Which is equal to

$$H(z) = \frac{p_z^2}{2m} + \frac{m}{2}\omega^2 \left(z - \frac{p_x}{m\omega}\right)^2 \tag{11}$$

It is exactly the equation of a harmonic oscillator with a center of equilibrium which moves depending on the x-position, even if it describes the movement in the z-direction.

3.1.3 Hamiltonian along the y-axis

As the movement along y-axis is not affected by the magnetic field, its Hamiltonian is just equal to p_y. When we develop $p_y = m v_y + \frac{e}{c} A_y$, we can simplify it by $p_y = m v_y$, as $A_y = 0$:

$$H(y) = \frac{p_y^2}{2m} \tag{12}$$

It means that the particle is just moving at its initial speed along the y-axis.

3.1.4 Global Hamiltonian

We can add the Hamiltonian (10), (11) and (12) respectively along the x, z and y-axis, since they are independent:

$$H(x,y,z) = \frac{p_x^2}{2m} + \frac{m}{2}\omega^2 \left(x + \frac{p_z}{m\omega}\right)^2 + \frac{p_z^2}{2m} + \frac{m}{2}\omega^2 \left(z - \frac{p_x}{m\omega}\right)^2 + \frac{p_y^2}{2m} \tag{13}$$

To conclude, we can recognize three parts:

- $\frac{p_x^2}{2m} + \frac{m}{2}\omega^2 \left(x + \frac{p_z}{m\omega}\right)^2$: harmonic oscillator centered around $x + \frac{p_z}{m\omega}$
- $\frac{p_z^2}{2m} + \frac{m}{2}\omega^2 \left(z - \frac{p_x}{m\omega}\right)^2$: harmonic oscillator centered around $z - \frac{p_x}{m\omega}$
- $\frac{p_y^2}{2m}$: free kinetic term, corresponding to motion unaffected by the magnetic field

3.2 Quantum Harmonic Oscillator

In this subsection, we determine the wave functions associated with the harmonic oscillators appearing in equation (13). We will first solve the problem on the dimension x, and then on the second dimension z. At the end, a wave function of the two-dimensional quantum harmonic oscillator will be given.

3.2.1 Dimension x

The Hamilton operator is defined as the sum of the kinetic and potential energies.

$$\hat{H} = T + V = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2(x + \frac{p_z}{m\omega})^2$$

Substituting $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$, we obtain:

$$\hat{H} = \frac{1}{2m}\left(i^2\hbar^2\frac{\partial^2}{\partial x^2}\right) + \frac{1}{2}m\omega^2(x + \frac{p_z}{m\omega})^2 = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2(x + \frac{p_z}{m\omega})^2$$

The time-dependent Schrödinger equation reads:

$$\hat{H}\Phi(x,t) = i\hbar\frac{\partial}{\partial t}\Phi(x,t)$$

Assuming a stationary state solution of the form

$$\Phi(x,t) = e^{-\frac{iEt}{\hbar}}\Phi(x),$$

and substituting it into the time-dependent equation yields the time-independent Schrödinger equation:

$$\hat{H}\Phi(x) = E\Phi(x) \tag{14}$$

To simplify the equation, we multiply both sides by $\frac{2}{\hbar\omega}$, and define the dimensionless variables $\gamma = \sqrt{\frac{m\omega}{\hbar}}(x + \frac{p_z}{m\omega})$ and $\tilde{E} = \frac{2E}{\hbar\omega}$ .This leads to:

$$\left[-\frac{\partial^2}{\partial\gamma^2} + \gamma^2\right]\Phi(\gamma) = \tilde{E}\Phi(\gamma)$$

We can subtract $\tilde{E}\Phi(\gamma)$ from both sides, in order to get the harmonic oscillator Schrödinger equation:

$$\left[-\frac{\partial^2}{\partial\gamma^2} + (\gamma^2 - \tilde{E})\right]\Phi(\gamma) = 0 \tag{15}$$

We may equivalently write the wave function in terms of γ:

$$\Phi(\gamma, t) = \Phi(\gamma)e^{-\frac{iEt}{\hbar}} \tag{16}$$

The detailed calculations are omitted here for clarity, but can be found in the Appendix 5.3.1 for the time-independent Schrödinger equation, and in the Appendix 5.3.2 for the harmonic oscillator Schrödinger.

Resolution of the harmonic oscillator Schrödinger

Let us now examine the asymptotic behavior of this equation as $\gamma \to +\infty$:

$$\lim_{\gamma \to +\infty} \left[-\frac{\partial^2}{\partial \gamma^2} + (\gamma^2 - \tilde{E}) \right] \Phi(\gamma) = \left[-\frac{\partial^2}{\partial \gamma^2} + \gamma^2 \right] \Phi(\gamma)$$

We can propose a solution of the form:

$$\Phi(\gamma) = Ae^{-\frac{\gamma^2}{2}} + Be^{\frac{\gamma^2}{2}}$$

which is an *ansatz* (i.e. a trial solution based on the structure of the differential equation). However, the second term diverges as $\gamma \to +\infty$, which is unacceptable since the wavefunction must be square-integrable:

$$\int_{-\infty}^{+\infty} |\Phi(\gamma)|^2 d\gamma = 1$$

To prevent divergence, we must impose $B = 0$. Hence, the asymptotic solution is:

$$\Phi(\gamma) \sim Ae^{-\frac{\gamma^2}{2}} \quad \text{as } \gamma \to +\infty$$

We now seek solutions that oscillate around the origin. These oscillations can be encoded in a function $h(\gamma)$, so we write:

$$\Phi(\gamma) = h(\gamma)e^{-\frac{\gamma^2}{2}} \tag{17}$$

Determination of $h(\gamma)$

We seek a differential equation for $h(\gamma)$. Differentiating $\Phi(\gamma)$ two times yields:

$$\begin{aligned}
\Phi'(\gamma) &= h'(\gamma)e^{-\frac{\gamma^2}{2}} - \gamma h(\gamma)e^{-\frac{\gamma^2}{2}} = \left[h'(\gamma) - \gamma h(\gamma)\right] e^{-\frac{\gamma^2}{2}} \\
\Phi''(\gamma) &= \left[h''(\gamma) - h(\gamma) - \gamma h'(\gamma)\right] e^{-\frac{\gamma^2}{2}} - \gamma \left[h'(\gamma) - \gamma h(\gamma)\right] e^{-\frac{\gamma^2}{2}} \\
&= \left[h''(\gamma) - h(\gamma) - \gamma h'(\gamma) - \gamma h'(\gamma) + \gamma^2 h(\gamma)\right] e^{-\frac{\gamma^2}{2}} \\
&= \left[h''(\gamma) - 2\gamma h'(\gamma) + (\gamma^2 - 1)h(\gamma)\right] e^{-\frac{\gamma^2}{2}}.
\end{aligned}$$

Substituting $\Phi(\gamma)$ in the harmonic oscillator Schrödinger defined in equation (15) yields:

$$\begin{aligned}
-\left[h''(\gamma) - 2\gamma h'(\gamma) + (\gamma^2 - 1)h(\gamma)\right] e^{-\frac{\gamma^2}{2}} + (\gamma^2 - \tilde{E})h(\gamma)e^{-\frac{\gamma^2}{2}} &= 0 \\
-\left[h''(\gamma) - 2\gamma h'(\gamma) + (\gamma^2 - 1)h(\gamma)\right] + (\gamma^2 - \tilde{E})h(\gamma) &= 0 \\
-h''(\gamma) + 2\gamma h'(\gamma) - (\gamma^2 - 1)h(\gamma) + (\gamma^2 - \tilde{E})h(\gamma) &= 0 \\
-h''(\gamma) + 2\gamma h'(\gamma) + h(\gamma) - \tilde{E}h(\gamma) &= 0 \\
-h''(\gamma) + 2\gamma h'(\gamma) + (1 - \tilde{E})h(\gamma) &= 0
\end{aligned}$$

Writing the equation with a positive coefficient for $h''(\gamma)$ is more convenient. Thus, we obtain:

$$h''(\gamma) - 2\gamma h'(\gamma) + (\tilde{E} - 1)h(\gamma) = 0 \tag{18}$$

We can find $h(\gamma)$ by introducing Hermite polynomials, which are known to be the solutions of the quantum harmonic oscillator equation. Nevertheless, we can also use Maclaurin series, which are Taylor series centered at the origin. Although tedious and long, they provide an approach that allows us to obtain the quantization of energy levels, which is a fundamental theorem in quantum mechanics. We will first use the Maclaurin series, and then introduce Hermite polynomials.

Time-independent wave function with Maclaurin series

In order to keep this part as clear as possible, we will write the Maclaurin series expansion in the Appendix 5.4. The harmonic oscillator Schrödinger equation defined in equation (15) is satisfied by $\Phi(\gamma) = h(\gamma)e^{-\frac{\gamma^2}{2}}$ with $h(\gamma) = \sum_{n=0}^{+\infty} a_n \gamma^n$, provided that:

1. The coefficients a_n satisfy the recurrence relation:

$$a_{n+2} = \frac{2n + 1 - \tilde{E}}{(n+1)(n+2)}\, a_n$$

2. The series terminates, which imposes the quantization condition:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

The quantization condition is fundamental in quantum mechanics, as it shows that the energy levels are discrete and can only take specific values.

Time-independent wave function with Hermite polynomials

Hermite's polynomials also satisfy the equation (18), they are defined as:

$$H_n(\gamma) = (-1)^n e^{\gamma^2} \frac{d^n}{d\gamma^n}\, e^{-\gamma^2}$$

For a proof of why Hermite's polynomials are a solution, see Appendix 5.5. Thus, we obtain a wave function in terms of γ and t

$$\Phi_n(\gamma) = N_n H_n(\gamma) e^{-\frac{\gamma^2}{2}}$$

Note that we introduced N_n as a normalization factor, so that the wave function is square-integrable. It is defined as $N_n = \frac{1}{2^{n/2}\sqrt{\frac{n!}{\sqrt{\pi}}}}$.

The last paragraph leads to the same wave function, but the Hermite polynomials method is considered as the standard method for solving the quantum harmonic oscillator. Thus, we will continue our reasoning with Hermite polynomials.

Time-dependent wave function

From the wave function in terms of γ defined in equation (16), we can write:

$$\Phi_n(\gamma, t) = \Phi_n(\gamma) e^{-\frac{iE_n t}{\hbar}} = N_n H_n(\gamma) e^{-\frac{\gamma^2}{2}} e^{-\frac{iEt}{\hbar}}$$

Since the dimensionless variable $\gamma = \frac{m\omega}{4\hbar} x$ depends on the space variable x, we can write $\Phi_n(x,t)$ instead:

$$\Phi_n(x,t) = N_n H_n(\gamma) e^{-\frac{\gamma^2}{2}} e^{-\frac{iE_n t}{\hbar}} \tag{19}$$

Probability density

Unlike classical mechanics, in quantum mechanics, a particle is not located at a precise point. Instead, the particle's position is represented as the absolute square of its wave function $\Phi(x,t)$. This means that $|\Psi(x,t)|^2 dx$ represents the probability of finding the particle between x and $x + dx$. Consequently, the particle's location is uncertain. The probability density gives the probability of finding the particle between two points, a and b, and is defined as:

$$P(a \leq x \leq b) = \int_a^b |\Psi_n(x,t)|^2 dx$$

When substituting the wave function of the one-dimensional quantum harmonic oscillator defined in equation (19) in the probability density formula, we find:

$$\begin{aligned}
P(a \leq x \leq b) &= \int_a^b |\Psi_n(x,t)|^2 \, dx = \int_a^b \left|\Psi_n(x) e^{-\frac{iE_n t}{\hbar}}\right|^2 dx \\
&= \int_a^b |\Psi_n(x)|^2 \left|e^{-\frac{iE_n t}{\hbar}}\right|^2 dx = \int_a^b |\Psi_n(x)|^2 \, e^{-\frac{iE_n t}{\hbar}} e^{\frac{iE_n t}{\hbar}} \, dx \\
&= \int_a^b |\Psi_n(x)|^2 \, dx
\end{aligned}$$

This formulation will be particularly useful when extending the method to the two-dimensional harmonic oscillator, as discussed in the next section.

3.2.2 Dimension z

The wave function along the z-axis is the same as the one along the x-axis, except for the variable γ. In the previous section, we defined γ as $\sqrt{\frac{m\omega}{\hbar}}(x + \frac{p_z}{m\omega})$. We will now define γ as $\sqrt{\frac{m\omega}{\hbar}}(z - \frac{p_x}{m\omega})$ since the equilibrium position has moved. Thus, we obtain the following wave function for the quantum harmonic oscillator in the z dimension:

$$\Phi_n(z,t) = N_n H_n(\gamma) e^{-\frac{\gamma^2}{2}} e^{-\frac{iE_n t}{\hbar}} \tag{20}$$

with $\gamma = \sqrt{\frac{m\omega}{\hbar}}(z - \frac{p_z}{m\omega})$ and $E_n = \hbar\omega(n + \frac{1}{2})$.

3.2.3 Two Dimensions

The Hamiltonian contains two independent one-dimensional harmonic oscillators. They can be expressed as a sum, so the full Hamiltonian can be written as a two-dimensional harmonic oscillator. In this way, the wave function of the two-dimensional harmonic oscillator is a product of the harmonic oscillator in dimension x, and the one in dimension z.

$$H(x,z) = \frac{p_z^2}{2m} + \frac{m}{2}\omega^2(z - \frac{p_x}{m\omega})^2 + \frac{p_x^2}{2m} + \frac{m}{2}\omega^2(x + \frac{p_z}{m\omega})^2$$

By factorizing both $\frac{1}{2m}$ and $\frac{m}{2}\omega^2$, we find:

$$H(x,z) = \frac{p_z^2 + p_x^2}{2m} + \frac{m}{2}\omega^2\left[(z - \frac{p_x}{m\omega})^2 + (x + \frac{p_z}{m\omega})^2\right] \tag{21}$$

Wave function

In the previous subsection, we found these wave functions using Hermite polynomials:

$$\Phi_{n_x}(x,t) = N_{n_x} H_{n_x}(\gamma_x) e^{-\frac{\gamma_x^2}{2}} e^{-\frac{iE_{n_x} t}{\hbar}}$$
$$\Phi_{n_z}(z,t) = N_{n_z} H_{n_z}(\gamma_z) e^{-\frac{\gamma_z^2}{2}} e^{-\frac{iE_{n_z} t}{\hbar}}$$

with $\gamma_x = \sqrt{\frac{m\omega}{\hbar}}(x + \frac{p_z}{m\omega})$ and $\gamma_z = \sqrt{\frac{m\omega}{\hbar}}(z - \frac{p_x}{m\omega})$.

Multiplying the wave functions, we obtain:

$$\begin{aligned}\Phi_{n_x,n_z}(x,z,t) &= \Phi_{n_x}(x) e^{\frac{-iE_{n_x} t}{\hbar}} \Phi_{n_z}(z)_n e^{\frac{-iE_{n_z} t}{\hbar}} \\ &= \Phi_{n_x}(x)\Phi_{n_z}(z) e^{\frac{-it}{\hbar}(E_{n_x} + E_{n_z})}\end{aligned}$$

Finally, the wave function of the two-dimensional harmonic oscillator is:

$$\Phi_{n_x,n_z}(x,z,t) = \Phi_{n_x}(x)\Phi_{n_z}(z) \tag{22}$$

This expression represents the wave function for the two-dimensional harmonic oscillator.

Energy

In order to get the energy levels, we can use the energy contribution of each dimension:

$$E_{n_x} = \frac{\hbar\omega}{2}\left(n_x + \frac{1}{2}\right), \qquad E_{n_z} = \frac{\hbar\omega}{2}\left(n_z + \frac{1}{2}\right)$$

The energy contribution of the two-dimensional harmonic oscillator is thus:

$$E_n = E_{n_x} + E_{n_z} = \hbar\omega\left(n_x + \frac{1}{2}\right) + \hbar\omega\left(n_z + \frac{1}{2}\right) = \hbar\omega\left(n_x + n_z + \frac{1}{2}\right)$$

Defining $n = n_x + n_z$ yields:

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) \tag{23}$$

Notice that the quantization condition has the same structure as the one-dimensional result, except that the quantum number n now comes from two independent dimensions. We will see later that n is the principal quantum number in the solution of $H(x, y, z)$.

Probability density

When substituting the wave function of the 2D quantum harmonic oscillator defined in equation (22) in the probability density formula, we find:

$$\begin{aligned} P(x_a \leq x \leq x_b,\, z_a \leq z \leq z_b) &= \int_{x_a}^{x_b}\int_{z_a}^{z_b} \left|\Phi_{n_x}(x)\Phi_{n_z}(z)e^{\frac{-it}{\hbar}(E_{n_x}+E_{n_z})}\right|^2 dz\,dx \\ &= \int_{x_a}^{x_b}\int_{z_a}^{z_b} \left|\Psi_{n_x}(x)\right|^2 \left|\Psi_{n_z}(z)\right|^2 dz\,dx. \end{aligned}$$

In the same way, the exponentials cancel out when taking the modulus squared, because

$$\left|e^{-i(E_{n_x}+E_{n_z})t/\hbar}\right|^2 = e^{-i(E_{n_x}+E_{n_z})t/\hbar} \cdot e^{i(E_{n_x}+E_{n_z})t/\hbar} = 1.$$

As in the one-dimensional harmonic oscillators, the probability density of the two-dimensional oscillator does not depend on time.

3.3 Kinetic energy along y

In this subsection, we determine the wave functions associated with the term $\frac{p_y^2}{2m}$ appearing in equation (13). We first substitute $\hat{H} = \frac{\hat{p}_y^2}{2m}$ in the time-independent Schrödinger equation:

$$\begin{aligned} \hat{H}\Phi(y) = E\Phi(y) &\Longleftrightarrow \frac{\hat{p}_y^2}{2m}\Phi(y) = E\Psi(y) \\ &\Longleftrightarrow -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Phi(y) = E\Psi(y) \\ &\Longleftrightarrow -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Phi(y) - E\Psi(y) = 0 \end{aligned}$$

We obtain a differential equation, for which we can find the function $\Psi(y)$ by making an *ansatz*:

$$\Psi(y) = e^{\lambda y}$$

Substituting the *ansatz* in the differential equation yields:

$$\begin{aligned}
&-\frac{\hbar^2}{2m}\lambda^2 e^{\lambda y} - E e^{\lambda y} = 0 \Longleftrightarrow e^{\lambda y}\left(-\frac{\hbar^2}{2m}\lambda^2 - E\right) = 0 \\
&\Longleftrightarrow -\frac{\hbar^2}{2m}\lambda^2 - E = 0
\end{aligned}$$

Solving this quadratic equation (with $\Delta = -4\frac{\hbar^2}{2m}E = -\frac{2\hbar^2 E}{m} < 0$) gives the solution:

$$\lambda = \pm i\,\frac{\sqrt{2mE}}{\hbar} = \pm i k_y \quad \text{with} \quad k_y = \frac{\sqrt{2mE}}{\hbar}$$

Thus, the wave function associated with $\hat{H} = \frac{\hat{p}_y^2}{2m}$ is:

$$\Psi_n(y,t) = e^{\pm i k_y y} e^{-\frac{iE_n t}{\hbar}} \tag{24}$$

with $k_y = \frac{\sqrt{2mE_n}}{\hbar}$ and $E_n = \frac{\hbar^2 k_y^2}{2m}$

3.4 Conclusion in quantum mechanics

We write the global wave function using the harmonic oscillators wave functions and the wave function corresponding to the movement in the y-direction.

$$\Phi_{n_x,n_y,n_z,k_y}(x,y,z,t) = \Phi_{n_x}(x)\Phi_{n_z}(z) e^{\pm i k_y y} e^{-\frac{iE_{n_y} t}{\hbar}} \tag{25}$$

where:

$$\begin{aligned}
\Phi_{n_x}(x,t) &= N_{n_x}\, H_{n_x}(\gamma_x) e^{-\gamma_x^2/2} e^{-iE_{n_x}t/\hbar} & \Phi_{n_z}(z,t) &= N_{n_z}\, H_{n_z}(\gamma_z) e^{-\gamma_z^2/2} e^{-iE_{n_z}t/\hbar} \\
\gamma_x &= \sqrt{\frac{m\omega}{\hbar}}\left(x + \frac{p_z}{m\omega}\right) & \gamma_z &= \sqrt{\frac{m\omega}{\hbar}}\left(z - \frac{p_x}{m\omega}\right) \\
E_{n_x} &= \hbar\omega\left(n_x + \tfrac{1}{2}\right) & E_{n_z} &= \hbar\omega\left(n_z + \tfrac{1}{2}\right) \\
N_n &= \frac{1}{2^{n/2}}\sqrt{\frac{1}{n!\sqrt{\pi}}} & H_n(\gamma) &= (-1)^n e^{\gamma^2}\frac{d^n}{d\gamma^n} e^{-\gamma^2} \\
k_y &= \frac{\sqrt{2mE_{n_y}}}{\hbar} & E_{n_y} &= \frac{\hbar^2 k_y^2}{2m}
\end{aligned}$$

4 Conclusion

4.1 Numerical calculations

4.1.1 Classical mechanics

A complete, visual model of the particle's trajectory in classical mechanics can be found on this website. You will be able to modify the initial parameters, and see in real time how each one modifies the trajectory. A static graph can be seen below, with the initial conditions $v_{x,0} = 3$, $v_{y,0} = 3$, $v_{z,0} = 3$, $\omega = 2$, $y_0 = 0$. The source code used to obtain this figure is publicly available at this address.

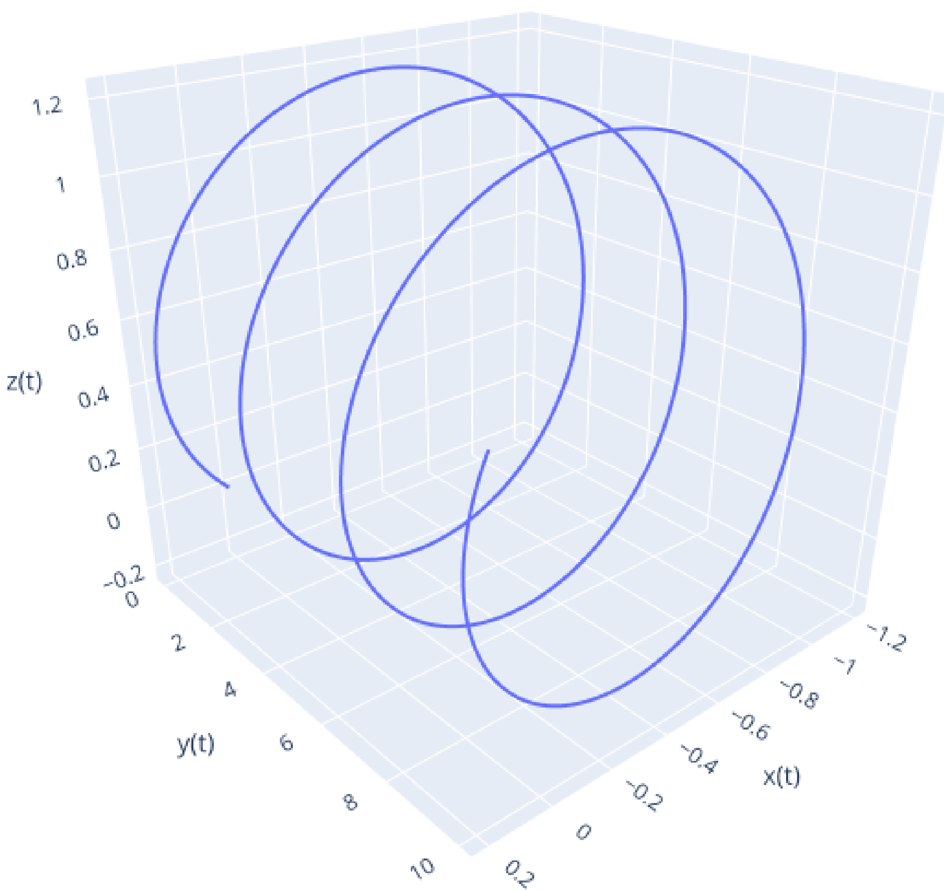
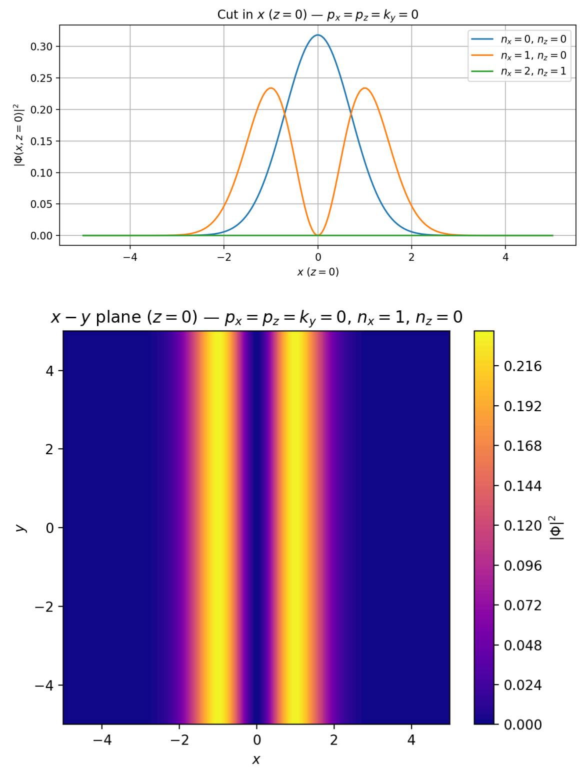


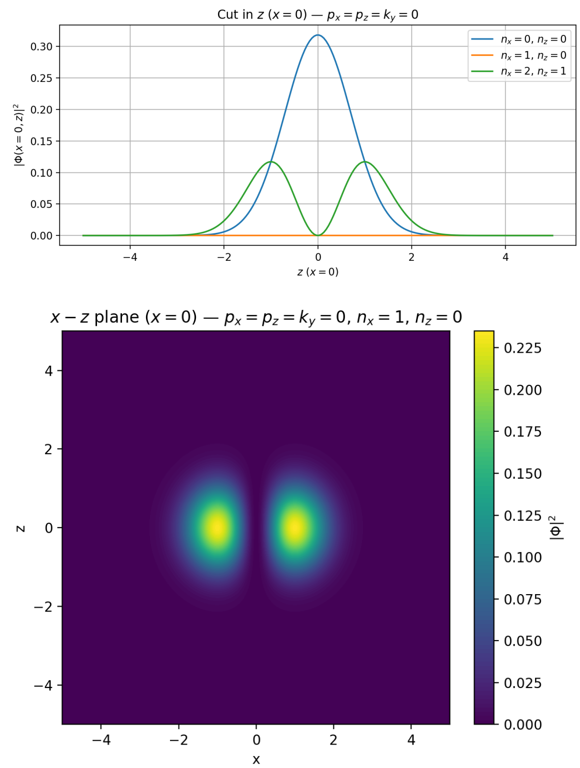
Figure 1: Visualization of $q(t)$.

4.1.2 Quantum mechanics

Thanks to our equation, we can determine the probability density of the particle in space, which allows us to see the fundamental difference between classical and quantum mechanics. We used the AI *Microsoft Copilot* to generate a Python code that outputs models. We initially tried to code it ourselves, but it was too complicated, that is why we needed the help of an AI to write this code. A lot more graphs can be obtained from the code. The source code used to obtain this figure is also publicly available at this address.



(a) view from the top



(b) view from in front

Figure 2: Different visualizations of the probability density.

We can clearly see that the path of the particle is made up of two "tubes" of probability. In the above visualization, we can observe the two locations where the probability is the highest. If we change the parameter, as the energy level in the dimensions x,y or z, it will change the shape of the probability location of the particle. We can observe other wave functions with the code we used.

5 Appendix

5.1 Acknowledgments

5.2 Detailed solution choice

We consider the harmonic oscillator equation:

$$\ddot{v}_x + \omega^2 v_x = 0$$

To solve it, we assume a trial solution of the form $v_x(t) = e^{rt}$, where $r \in \mathbb{C}$. This choice is motivated by the fact that:

- The derivatives of e^{rt} are proportional: $\frac{d^k}{dt^k} e^{rt} = r^k e^{rt}$.
- For equations with complex roots, like $r = \pm i\omega$, Euler's identity allows us to rewrite solutions with sines and cosines.

Substituting the trial solution into the differential equation yields:

$$r^2 e^{rt} + \omega^2 e^{rt} = 0 \quad \Longleftrightarrow \quad e^{rt}(r^2 + \omega^2) = 0$$

As $e^{rt} \neq 0$ for all $t \in \mathbb{R}$ (and even $\mathbb{C}$), we know that $r^2 + w^2 = 0$ and $r^2 = -\omega^2$. Solving in $\mathbb{C}$ gives

$$r = \pm\sqrt{-\omega^2} = \pm i\sqrt{w^2} = \pm iw$$

Thus, we consider the two possible values of r: $v_x(t) = C_1 e^{-i\omega t} + C_2 e^{i\omega t}$. Applying Euler's identity $e^{\pm i\omega} = \cos\omega \pm i\sin\omega$:

$$\begin{aligned}
v_x(t) &= C_1(\cos -\omega t + i\sin -\omega t) + C_2(\cos\omega t + i\sin\omega t) \\
&= C_1(\cos\omega t - i\sin\omega t) + C_2(\cos\omega t + i\sin\omega t) \\
&= C_1\cos\omega t - C_1 i\sin\omega t + C_2\cos\omega t + C_2 i\sin\omega t \\
&= (C_1 + C_2)\cos\omega t + (C_2 - C_1) i\sin\omega t
\end{aligned}$$

Letting $A = C_1 + C_2, \quad B = (C_2 - C_1)i$, we obtain the general solution:

$$v_x(t) = A\cos\omega t + B\sin\omega t$$

5.3 Harmonic oscillator

5.3.1 Time-independent Schrödinger

Substituting $\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2$ and $\Phi(x,t) = e^{-\frac{iEt}{\hbar}}\Phi(x)$ into the left-hand side of the time-dependent Schrödinger equation yields:

$$\begin{aligned}
-\frac{\hbar^2}{2m}\frac{\partial^2\Phi(x,t)}{\partial x^2} + \frac{1}{2}m\omega^2 x^2\Phi(x,t) &= -\frac{\hbar^2}{2m}\left[e^{-\frac{iEt}{\hbar}}\frac{\partial^2\Phi(x)}{\partial x^2}\right] + \frac{1}{2}m\omega^2 x^2 e^{-\frac{iEt}{\hbar}}\Phi(x) \\
&= \left[-\frac{\hbar^2}{2m}\frac{\partial^2\Phi(x)}{\partial x^2} + \frac{1}{2}m\omega^2 x^2\Phi(x)\right] e^{-\frac{iEt}{\hbar}} \\
&= \hat{H}\Phi(x)e^{-\frac{iEt}{\hbar}}
\end{aligned}$$

We can do the same in the right side of the Schrödinger equation:

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\Phi(x,t) &= i\hbar\frac{\partial}{\partial t}\left[\Phi(x)e^{-\frac{iEt}{\hbar}}\right] \\
&= i\hbar\left[-\frac{iE}{\hbar}e^{-\frac{iEt}{\hbar}}\Phi(x)\right] \\
&= -i^2E\Phi(x)e^{-\frac{iEt}{\hbar}} = E\Phi(x)e^{-\frac{iEt}{\hbar}}
\end{aligned}$$

Equating both sides and dividing by $e^{-\frac{iE}{\hbar}} \neq 0$ gives:

$$\hat{H}\Phi(x) = E\Phi(x)$$

5.3.2 Schrödinger equation for the quantum harmonic oscillator

We multiply both sides of the time-independent Schrödinger equation

$$\hat{H}\Phi(x) = E\Phi(x)$$

by the constant factor $\frac{2}{\hbar\omega}$:

$$\frac{2\hat{H}\Phi(x)}{\hbar\omega} = \frac{2E}{\hbar\omega}\Phi(x)$$

We now introduce the following dimensionless quantities $\tilde{E} = \frac{2E}{\hbar\omega}$ and $\gamma = \sqrt{\frac{m\omega}{\hbar}}x$ Substituting into the equation, we get:

$$\begin{aligned}
\left[-\frac{2}{\hbar\omega}\cdot\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{2}{\hbar\omega}\cdot\frac{1}{2}m\omega^2x^2\right]\Phi(x) &= \frac{2E}{\hbar\omega}\Phi(x) \\
\left[-\frac{\hbar}{m\omega}\frac{\partial^2}{\partial x^2} + \frac{m\omega}{\hbar}x^2\right]\Phi(x) &= \tilde{E}\Phi(x)
\end{aligned}$$

Next, we change variables from x to the dimensionless parameter γ. Since

$$\gamma = \sqrt{\frac{m\omega}{\hbar}}x \quad \Longleftrightarrow \quad x = \sqrt{\frac{\hbar}{m\omega}}\gamma,$$

we apply the chain rule to express derivatives with respect to x in terms of γ:

$$\frac{\partial\Phi(x)}{\partial x} = \frac{\partial\gamma}{\partial x}\frac{\partial\Phi(\gamma)}{\partial\gamma} = \sqrt{\frac{m\omega}{\hbar}}\frac{\partial\Phi(\gamma)}{\partial\gamma} \quad \Longrightarrow \quad \frac{\partial^2\Phi(x)}{\partial x^2} = \left(\frac{m\omega}{\hbar}\right)\frac{\partial^2\Phi(\gamma)}{\partial\gamma^2}$$

Substituting into the differential equation:

$$-\frac{\hbar}{m\omega}\frac{\partial^2\Phi(x)}{\partial x^2} = -\frac{\hbar}{m\omega}\frac{m\omega}{\hbar}\frac{\partial^2}{\partial\gamma^2}\Phi(\gamma) = -\frac{\partial^2\Phi(\gamma)}{\partial\gamma^2}$$

Thus, the Schrödinger equation becomes:

$$\left[-\frac{\partial^2}{\partial\gamma^2} + \gamma^2\right]\Phi(\gamma) = \tilde{E}\Phi(\gamma)$$

5.4 Maclaurin series expansion

The Maclaurin series is a special case of the Taylor series centered at the origin (x=0). It offers a way to represent a function as an infinite sum of polynomial terms. We expand $h(\gamma)$ in a Maclaurin series to solve the differential equation:

$$
\begin{aligned}
h(\gamma) &= \sum_{n=0}^{+\infty} a_n \gamma^n \quad \text{with} \quad a_n = \frac{f^{(n)}(0)}{n!} \\
h'(\gamma) &= \sum_{n=0}^{+\infty} n a_n \gamma^{n-1} \\
h''(\gamma) &= \sum_{n=0}^{+\infty} n(n-1) a_n \gamma^{n-2}
\end{aligned}
$$

Substituting $h''(\gamma)$, $h'(\gamma)$ and $h(\gamma)$ in the differential equation (18) yields:

$$\sum_{n=0}^{+\infty} n(n-1) a_n \gamma^{n-2} - 2\gamma \sum_{n=0}^{+\infty} n a_n \gamma^{n-1} + (\tilde{E}-1) \sum_{n=0}^{+\infty} a_n \gamma^n = 0$$

To combine these sums, all series must have the same power of γ.

Index shifting

For the first term $h''(\gamma)$, set $k = n-2 \iff n = k+2$:

$$h''(\gamma) = \sum_{n=0}^{+\infty} n(n-1) a_n \gamma^{n-2} = \sum_{k=0}^{+\infty} (k+2)(k+1) a_{k+2} \gamma^k = \sum_{n=0}^{+\infty} (n+2)(n+1) a_{n+2} \gamma^n$$

For the second term $-2\gamma h'(\gamma)$:

$$-2\gamma h'(\gamma) = -2\gamma \sum_{n=0}^{+\infty} n a_n \gamma^{n-1} = \sum_{n=0}^{+\infty} -2n a_n \gamma^n$$

Combining all terms:

$$\sum_{n=0}^{+\infty} \Big[(n+1)(n+2) a_{n+2} - 2n a_n + (\tilde{E}-1) a_n \Big] \gamma^n = 0$$

This implies:

$$
\begin{aligned}
(n+1)(n+2) a_{n+2} - 2n a_n + (\tilde{E}-1) a_n &= 0 \\
(n+1)(n+2) a_{n+2} - (2n - \tilde{E} - 1) a_n &= 0 \\
a_{n+2} &= \frac{2n - \tilde{E} + 1}{(n+1)(n+2)} a_n
\end{aligned}
$$

As $\Phi(\gamma) = h(\gamma)\, e^{-\frac{\gamma^2}{2}}$ and $\Phi \in L^2(\mathbb{R})$, we require that $h(\gamma)$ does not grow exponentially as $\gamma \to \infty$. This condition is satisfied if the series terminates. Let $n = n_{\max}$ such that $a_{n_{\max}+2} = 0$.

From $n = n_{max}$, $a_{n_{max}+2} = 0$:

$$\begin{aligned} a_{n_{max}+2} = \frac{2n_{max} - \tilde{E} + 1}{(n_{max}+1)(n_{max}+2)} a_{n_{max}} &= 0 \\ 2n_{max} - \tilde{E} + 1 &= 0 \\ \tilde{E} &= 2n_{max} + 1 \end{aligned}$$

Since $\tilde{E} = \frac{2E}{\hbar\omega}$, the quantized energy levels are:

$$E = \hbar\omega\left(n_{max} + \frac{1}{2}\right) \quad \text{with} \quad n_{max} \in \mathbb{N}$$

5.5 Hermite's polynomials

We will prove that Hermite's polynomials are a solution to the equation (18):

$$h''(\gamma) - 2\gamma h'(\gamma) + (\tilde{E} - 1)h(\gamma) = 0$$

Hermite's polynomials are defined as follow:

$$H_n(\gamma) = (-1)^n e^{\gamma^2} \frac{d^n}{d\gamma^n} e^{-\gamma^2}$$

Firstly, we will develop the term $(\tilde{E} - 1)$ by remembering that

$$\tilde{E} - 1 = \frac{2E}{\hbar\omega} - 1 = \frac{2\hbar\omega(n + \frac{1}{2})}{\hbar\omega} - 1 = 2n + 1 - 1 = 2n$$

Incorporating this term in the equation, it gives us

$$h''(\gamma) - 2\gamma h'(\gamma) + 2nh(\gamma) = 0$$

Thus, we need to prove Hermite's polynomials are a solution for $n \in \mathbb{N}$. We will use a proof by recurrence. We know that :

$$H_{n+1}(\gamma) = 2\gamma H_n - 2nH_{n-1}$$
$$H_n'(\gamma) = 2nH_{n-1}(\gamma)$$

We want to prove that, for all $n \in N$,

$$H_n''(\gamma) - 2\gamma H_n'(\gamma) + 2nH_n(\gamma) = 0 \tag{26}$$

Firstly, we will initialize:

$$H_0(\gamma) = 1 \implies H_0''(\gamma) - 2\gamma H_0'(\gamma) + 0H_0(\gamma) = 0$$

$$H_1(\gamma) = 2\gamma \to H_1''(\gamma) - 2\gamma H_1'(\gamma) + 2H_1(\gamma) = -4\gamma + 4\gamma = 0$$

Assuming that Hermite's polynomial is a solution of the equation for a certain $k \in N$, we want to prove that it is also a solution for $k + 1$.

$$H''(\gamma)_{k+1} - 2\gamma H'(\gamma)_{k+1} + 2nH(\gamma)_{k+1} = 0$$

We calculate

$$H''_{k+1}(\gamma) = 2(k+1)H'_k(\gamma) = 4k(k+1)H_{k-1}(\gamma) \tag{27}$$

and

$$-2\gamma H'(\gamma)_{k+1} = -4\gamma(k+1)H(\gamma)_k \tag{28}$$

We replace, so it gives us :

$$4k(k+1)H_{k-1}(\gamma) - 4\gamma(k+1)H(\gamma)_k + 2(k+1)H_{k+1}(\gamma) \tag{29}$$

Thus, as $H_{k+1} = 2\gamma H_k(\gamma) - 2kH_{k-1}(\gamma)$, we find that

$$4k(k+1)H_{k-1}(\gamma) - 4\gamma(k+1)H(\gamma)_k + 2(k+1)(2\gamma H_k(\gamma) - 2kH_{k-1}) = 0 \tag{30}$$

So the Hermite's polynomial is a solution of the differential equation (18). Thus, it is an appropriate function to describe the quantum harmonic oscillators wave function. Furthermore, they are orthogonal to each other:

$$\int_{-\infty}^{+\infty} H_n H_m e^{-\gamma^2} d\gamma = 2^n n! \sqrt{\pi} \delta_{n,m}$$

where $\delta_{n,m}$ is the Kronecker's symbol, so they can form a base for an adequate Hilbert's space of wave function.

5.6 Factorization of the wave function

We have the global Hamiltonian (13). We know that it can be separated in three others Hamiltonian, as it is an addition of Hamiltonian in the defined and separated dimensions.

$$H = A + B + C$$

With A the Hamiltonian in the x-direction, B the Hamiltonian in the z-direction and C the Hamiltonian in the y-direction. As the three of them are independent Hamiltonian, each of them has a specific wave function of its own dimension:

$$\begin{aligned} A\psi(a) &= \alpha\psi(a) \\ B\psi(b) &= \beta\psi(b) \\ C\psi(c) &= \epsilon\psi(c) \end{aligned}$$

Now,

$$H\psi(a,b,c) = (A+B+C)\psi(a,b,c)$$

A way to obtain the previous triplets of equation with this equation is to have a global wave function that is a multiplication of all wave functions of dimension a, b and c. It gives us:

$$\psi(a,b,c) = \psi(a)\psi(b)\psi(c)$$

Thus:

$$\begin{aligned} (A+B+C)\psi(a,b,c) =& A\psi(a,b,c) + B\psi(a,b,c) + C\psi(a,b,c) \\ =& A\psi(a)\psi(b)\psi(c) + B\psi(a)\psi(b)\psi(c) + C\psi(a)\psi(b)\psi(c) \end{aligned}$$

However, as each Hamiltonian affects only the wave function of its own dimension, it gives the final result

$$
\begin{aligned}
H\psi(a,b,c) =& A\psi(a) + B\psi(b) + C\psi(c) \\
=& \alpha\psi(a) + \beta\psi(b) + \epsilon\psi(c)
\end{aligned}
$$

Since it respects the triplets of each dimensional equation, the whole wave function can therefore be written as the product of each dimensional wave function.